

Algorithm and Data Structure

TD Article de Recherche : Dynamic Time Warping (DTW)

Objectives

Read and understand a research article
Prepare an answer for a research question
Present as a group the answer

The research article that you will be reading is titled “*Dynamic Programming Algorithm Optimization for Spoken Word Recognition*”¹ that presents an algorithm for aligning sequences.

Question 1 : Explain how Dynamic Programming is used in this paper to compare two sequences. Ignore speech and focus only on the algorithm. What are the subproblems, the recurrence relation, and the role of the table?

Question 2 : Analyze the time and space complexity of the Dynamic Time Warping (DTW) algorithm in the following four cases :

1. Standard DTW (no constraints)
2. DTW with a window constraint
3. DTW with a slope constraint
4. DTW with both window and slope constraints

Explain clearly which parameters affect the asymptotic complexity and why.

Question 3 : Describe the recurrence relation used in Dynamic Time Warping (DTW). How do the window constraint and the slope constraint change the set of allowed transitions in the dynamic programming table?

Question 4 : Dynamic Time Warping (DTW) is usually presented using a two-dimensional dynamic programming table. However, not all of this table is always necessary.

1. Explain which previously computed values are required to compute a new DP cell in DTW.
2. Using this observation, describe how the space complexity of DTW can be reduced.
3. What information is lost when this space optimization is applied, and in which situations is this tradeoff acceptable?

Question 5 : DTW computes a final distance using a DP table, but often we also want the actual alignment (the warping path). Explain how to reconstruct the warping path from the DP table.

1. What extra information (if any) must be stored while computing the DP table?
2. Give pseudocode for backtracking to produce the warping path from (I, J) back to $(1, 1)$.
3. How does using a window constraint and/or slope constraint affect path reconstruction (if at all)?

Question 6 : Dynamic Time Warping (DTW) compares two sequences by repeatedly using a local distance function $d(x, y)$, which is summed along a warping path.

1. Based on how the distance function is used in the DTW recurrence in the paper, what basic assumptions must hold for this distance function so that the algorithm behaves sensibly?

1. <https://jeffe.cs.illinois.edu/teaching/compgeom/refs/Sakoe-Chiba-DTW.pdf>

2. Explain how changing the distance function can change the final DTW alignment and distance, even though the dynamic programming structure stays the same.
3. Give one example of a distance function mentioned in the paper or a simple alternative, and explain how it might influence the alignment.

Question 7 : The paper describes two variants of Dynamic Time Warping : symmetric and asymmetric formulations.

1. Describe, at an algorithmic level, the difference between symmetric and asymmetric DTW.
2. Explain how this difference affects the way costs are accumulated in the dynamic programming algorithm.
3. Why might one formulation give better results than the other, even though both use dynamic programming?

Question 8 : Dynamic Time Warping is designed to align sequences that may differ in speed or timing. However, it does not always produce meaningful results.

1. Give one example of a situation where DTW may produce a poor or misleading alignment.
2. Explain, using the DTW algorithm and its constraints, why this failure happens.
3. Describe how one of the constraints discussed in the paper (window constraint or slope constraint) can help reduce this problem.